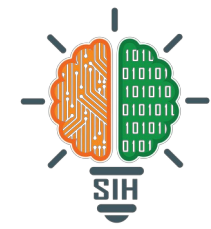


SMART INDIA HACKATHON 2026



SMART INDIA
HACKATHON
2026

TITLE PAGE



VisionBridge

NATIVE ANDROID ACCESSIBILITY COMPANION



Now with **VisionBridge MedSafe** · accessible medicine-pack reading

An AI companion that helps people with low vision read, understand, move and stay safe — on the Android phone they already own.

When the AI is unsure, it says so and can connect a real person.

Say “Vision...” in English · हिन्दी · मराठी | Android 7.0 and above | no extra hardware

PROBLEM STATEMENT ID

26198

PROBLEM STATEMENT TITLE

Student Innovation

THEME

MedTech / BioTech / HealthTech

PS CATEGORY

Software

TEAM ID

135775

TEAM NAME

Prime Innovators



READ

Printed text · medicine packs



UNDERSTAND

Scenes · objects · money · live AI



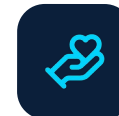
MOVE

Location · buses · obstacles ahead



STAY SAFE

SOS alert · medicine reminders



HUMAN HELP

Live video with a volunteer



IDEA / PROPOSED SOLUTION



VisionBridge — say “Vision...”, point the phone, hear the answer. When the AI is unsure, a real person can help.

THE PROBLEM • ONE DAY, MANY VISUAL BARRIERS

08:00  **Medicine strip**
Name, dose and expiry too small to read

09:30  **Bus stop**
Which bus is this? Where is it going?

11:00  **Footpath**
A step or an obstacle just ahead

15:00  **Shop counter**
Which note did I get back?

Any time  **“Where am I?”**
Help depends on someone being free

81%

could not access medicine names and doses under simulated moderate vision loss
Merenda et al., 2025 • reference [3]

THE SOLUTION • ONE VOICE-FIRST COMPANION FOR FIVE EVERYDAY JOBS

 **READ**
Hear printed words

Smart Reading

MedSafe

NEW
Reads medicine packs • see slides 3–5

 **UNDERSTAND**
Know what is around

Surroundings

Object Finder

Currency Reader

VisionBridge Live

AI Voice Call

 **MOVE**
Find your way

Where Am I?

Public Transport

Obstacle check

 **STAY SAFE**
Get help fast

Emergency SOS

Medicine reminders

 **HUMAN HELP**
Reach a real person

Volunteer video

Phone Assistant

Also on the home screen: radio, audiobooks and audio games • start talking with “Vision”, a two-finger double-tap or a shake (optional)

WHAT IS NEW • WHY IT IS DIFFERENT FROM A GENERIC AI APP

 **Says “I’m not sure”**

Photo answers are scored; below the threshold the app says it is unsure.

 **A person, one step away**

When the AI is unsure, a sighted volunteer can join on live video.

 **MedSafe for medicine packs**

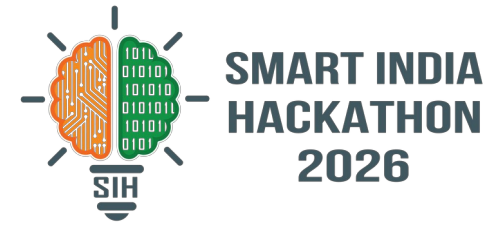
Reads only what is printed on the pack — directions word for word, no medical advice.

 **Voice-first, 3 languages**

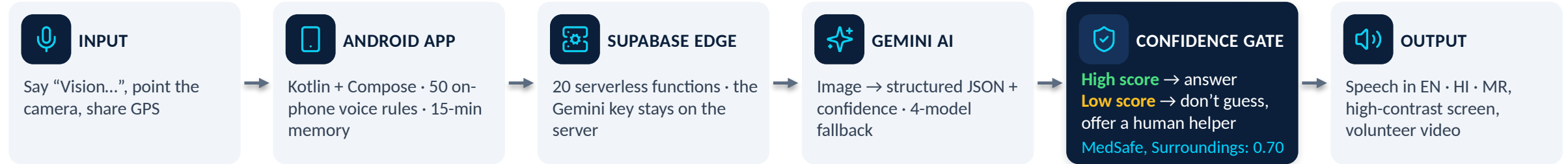
English, हिन्दी, मराठी; 50 common commands are matched on the phone.



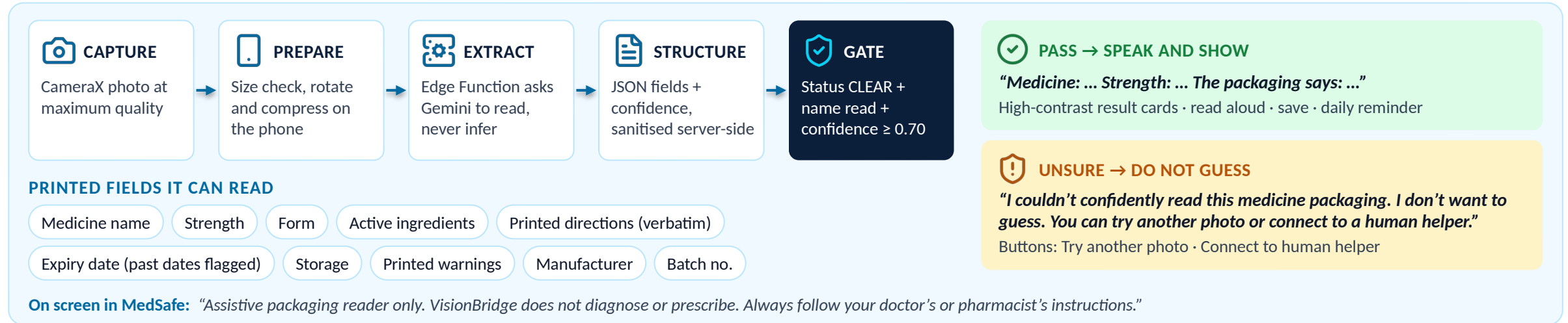
TECHNICAL APPROACH



HOW A PHOTO REQUEST WORKS · FROM A SPOKEN SENTENCE TO A SPOKEN ANSWER



MEDSAFE · THE STRICTEST GATE, APPLIED TO MEDICINE PACKAGING

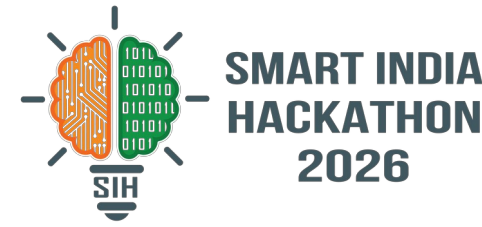


TECHNOLOGY ACTUALLY USED · CHECKED AGAINST THE CURRENT BUILD

ON THE PHONE	Kotlin	Jetpack Compose	CameraX	SpeechRecognizer	TextToSpeech	Fused Location	Media3 / ExoPlayer	SQLite + AlarmManager	minSdk 24
CLOUD BACKEND	Supabase Auth (JWT + silent refresh) · Postgres + row-level security · Edge Functions (Deno / TypeScript) · Supabase Realtime (call signalling)								
AI & CALLS	Gemini API → JSON + confidence · Gemini Live API (ephemeral tokens) · WebRTC video · OpenStreetMap Nominatim + Overpass · WhatsApp Cloud API (SOS)								



FEASIBILITY AND VIABILITY



FEASIBLE BECAUSE IT ALREADY RUNS • EVIDENCE FROM THE CURRENT BUILD

93 / 93

unit tests passing — latest local run, 14 Sep 2026

PASSED

End-to-end MedSafe test on a real phone, live backend + Gemini

EN • HI • MR

570 UI strings each; replies are spoken in the chosen language

20

Supabase Edge Functions called by the app — AI, SOS, location, audio

Android 7.0+

minSdk 24 — no extra device to buy, carry or charge

RISKS WE ACTUALLY HAVE → WHAT THE BUILD ALREADY DOES



Wrong AI answer



Photo results are scored. MedSafe and Surroundings withhold answers below 0.70; Currency and Transport offer a volunteer below 0.65.



Medicine over-trust



Printed text only — no diagnosis, dosing or suitability advice. Disclaimer in 3 languages; expired packs → “consult a pharmacist”.



AI quota or outage



Four Gemini models are tried in order within one request; the user hears a clear error and can retry.



No internet



Stated limit: AI features need a connection. Saved medicines and reminder alarms are stored on the phone and keep working offline.



Volunteer trust



Volunteers self-register today — a verified network is future scope. The video call starts only when a volunteer accepts.

PRIVACY AND SECURITY ALREADY IN PLACE



Gemini API key kept only in server secrets



Gemini Live runs on short-lived ephemeral tokens



Row-level security on all 13 tables (44 policies)



Users cannot promote themselves to admin



Medicine reminders stay private on the lock screen



Volunteer sees the user's camera; theirs stays off

VIABLE: runs on phones people already own • managed, pay-as-you-go cloud • modular — MedSafe was added as its own screens, Edge Function and tables without changing other assistants



IMPLEMENTED TODAY • IN THE ANDROID BUILD

- All assistants on slide 2, incl. MedSafe
- Gemini Live: real-time camera + voice
- SOS with WhatsApp location alert
- Confidence gate + volunteer video
- Voice and UI in EN • HI • MR
- Offline medicine list + reminder alarms

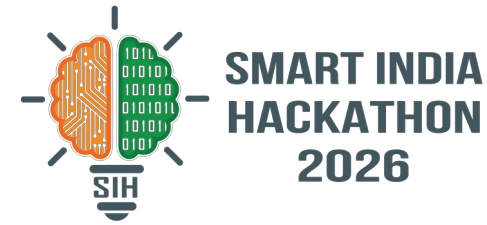


FUTURE SCOPE • NOT BUILT YET

- Offline obstacle warnings on the phone
- Verified volunteer network
- Field study with low-vision users
- Continuous hazard alerts while walking
- More Indian languages
- Pharmacist-reviewed MedSafe accuracy tests



IMPACT AND BENEFITS



WHO BENEFITS

People with low vision or blindness who use Android phones — plus their families, caregivers and the volunteers who help remotely.

2.2 billion+

people worldwide have a near or distance vision impairment • WHO, 2026 [1]

26.68%

of people aged 50+ in India have visual impairment • national survey, 2015–19 [2]

ONE REAL MOMENT • 8:30 PM, AT HOME, A MEDICINE STRIP IN HAND

⊗ WITHOUT VISIONBRIDGE

- The print is too small to read
- Waits for family — or guesses
- Expiry and directions stay unknown
- Dose times rely on memory or family



✓ WITH VISIONBRIDGE MEDSAFE

1



Asks by voice

“Vision, check my medicine.”
MedSafe opens — in English, Hindi or Marathi.

2



Hears the pack

Name, strength, printed directions and expiry are read out; past dates are flagged.

3



Saves and schedules

Adds it to My Medicines with a 9:00 PM daily reminder that works offline.

4



Never a guess

Blurry photo? “I don’t want to guess.” Try again or connect a volunteer.

WHAT CHANGES • SOCIAL, HEALTH, SAFETY, ECONOMIC AND ENVIRONMENTAL BENEFITS

SOCIAL



Independence

Read, travel and check money without waiting

SOCIAL



Inclusion

Voice-first in 3 languages, high-contrast UI

HEALTH



Safer medicine routine

Printed info + reminders — never advice

SAFETY



Human safety net

Volunteer video and SOS with live location

ECONOMIC



Low cost

Runs on the Android phone the user already has

ENVIRONMENT



No new device

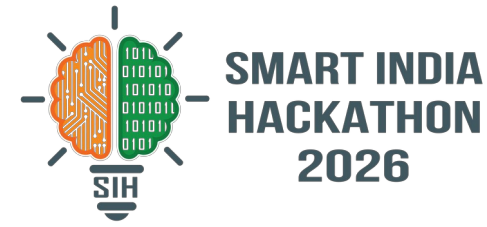
Nothing extra to make, charge or throw away

⊗ **WE DO NOT CLAIM:** clinical validation or regulatory approval • diagnosis, prescribing or dosing advice • guaranteed obstacle detection • accuracy or user-count figures • replacing a doctor, pharmacist, cane or guide.

Read what is printed. Don't guess what isn't.



RESEARCH AND REFERENCES



SEE IT • TRY IT • VERIFY IT

VisionBridge • Team Prime Innovators • Team ID 135775 • PS ID 26198



Demo video

Watch VisionBridge in action

youtu.be/ok7P_i5SeR4



Android source code

Kotlin app + Supabase backend

[github.com/shailesh-pande01/
visionbridge-app](https://github.com/shailesh-pande01/visionbridge-app)



Earlier web prototype

Browser version built before the
Android app

visionbridge-ai.vercel.app



Scan to watch
the demo video
or click the code

RESEARCH BEHIND THE IDEA

1

2.2 billion+ people live with a near or distance vision impairment.

World Health Organization — “Blindness and vision impairment” fact sheet, updated 10 Feb 2026.

who.int/news-room/fact-sheets/detail/blindness-and-visual-impairment

2

India, age 50+: visual impairment 26.68%, blindness 1.99%.

Vashist P. et al. National Blindness & Visual Impairment Survey 2015–19. PLOS ONE, 2022.

doi.org/10.1371/journal.pone.0271736

3

Simulated moderate vision loss: 81% could not access medicine names and doses; 75% missed the full expiry date.

Merenda T., Depasse F., Patris S. Explor. Res. Clin. Soc. Pharm. 18:100581, 2025.

doi.org/10.1016/j.rcsop.2025.100581

4

Blindness and low vision are specified disabilities under Indian law.

Rights of Persons with Disabilities Act, 2016 — Dept. of Empowerment of Persons with Disabilities.

depwd.gov.in/en/acts

OFFICIAL DOCUMENTATION FOR THE COMPONENTS USED

All links checked on 16 Sep 2026

ANDROID

- [CameraX](#)
- [TextToSpeech](#)
- [Scheduling alarms](#)
- [SpeechRecognizer](#)
- [Jetpack Compose](#)

GEMINI AI

- [Structured JSON output](#)
- [Live API](#)
- [Ephemeral tokens](#)

BACKEND AND SERVICES

- [Supabase row-level security](#)
- [Realtime](#)
- [Nominatim reverse geocoding](#)
- [Edge Functions](#)
- [WebRTC](#)
- [WhatsApp Cloud API](#)